

Name: _____ Date: _____

Period: _____

1. For each rational function, state which form (**factored** or **standard**) is more useful for the requested task, then answer the question.

(a) $r(x) = \frac{x^2 + 3x - 10}{x^2 - 4}$. Find all zeros, holes, and VAs.

Best form: **factored**

Factor numerator: $(x + 5)(x - 2)$ Factor denominator: $(x - 2)(x + 2)$

Zeros: $x = -5$ Holes: $(2, \frac{7}{4})$ VAs: $x = -2$

(b) $s(x) = \frac{3x^5 - 2x^2 + 1}{4x^3 - x}$. Describe the end behavior.

Best form: **standard**

As $x \rightarrow +\infty$: $s(x) \rightarrow +\infty$ As $x \rightarrow -\infty$: $s(x) \rightarrow +\infty$

2. Perform polynomial long division. Show all steps.

Divide $f(x) = 3x^3 - 5x^2 + 2x - 4$ by $g(x) = x - 2$.

Work space: $3x^3 \div x = 3x^2$; $3x^2(x - 2) = 3x^3 - 6x^2$; subtract $\rightarrow x^2 + 2x - 4$; $x^2 \div x = x$; $x(x - 2) = x^2 - 2x$; subtract $\rightarrow 4x - 4$; $4x \div x = 4$; $4(x - 2) = 4x - 8$; subtract $\rightarrow 4$.

$q(x) = 3x^2 + x + 4$ $r = 4$

Write as $f(x) = g(x) \cdot q(x) + r$: $(x - 2)(3x^2 + x + 4) + 4$

3. A rational function is defined by $R(x) = \frac{x^2 + x - 6}{x - 1}$.

- (a) Perform polynomial long division.

$R(x) = x + 2 + \frac{-4}{x - 1}$

- (b) State the equation of the slant asymptote. Explain in one sentence why this is the slant asymptote.

Slant asymptote: $y = x + 2$

As $x \rightarrow \pm\infty$, the remainder term $\frac{-4}{x-1} \rightarrow 0$, so $R(x)$ approaches the quotient line $y = x + 2$

- (c) What is the remainder when $x^2 + x - 6$ is divided by $x - 1$? What does this tell you about $R(1)$?

Remainder: -4 $R(1) = \text{undefined } (x = 1 \text{ is not in the domain})$

4. Use Pascal's Triangle to expand each expression. Show the row of Pascal's Triangle you used.

(a) $(x + 3)^3$

Row used: **Row 3: 1 3 3 1**

$(x + 3)^3 = x^3 + 9x^2 + 27x + 27$

(b) $(2x - 1)^4$ (Hint: let $a = 2x$ and $b = -1$)

Row used: **Row 4: 1 4 6 4 1**

$(2x - 1)^4 = 16x^4 - 32x^3 + 24x^2 - 8x + 1$

5. (AP-Style Multiple Choice) Which of the following statements about $p(x) = \frac{x^3 - 8}{x - 2}$ is correct?

- (A) p has a vertical asymptote at $x = 2$.
- (B) p has a hole at $x = 2$ with y -coordinate 12. ← correct
- (C) p simplifies to $x^2 - 2x + 4$ for all real x .
- (D) p has a slant asymptote of $y = x^2 + 2x + 4$.

Answer: **B** Show your reasoning:

Factor: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. **Shared factor $(x - 2)$ in numerator and denominator with equal multiplicity, so $x = 2$ is a hole. Substitute $x = 2$ into $x^2 + 2x + 4$: $4 + 4 + 4 = 12$. Hole at $(2, 12)$.**.....

Teacher Note

1(a): Hole at $x = 2$: substitute into simplified $\frac{x+5}{x+2}$ to get $\frac{7}{4}$.

1(b): Leading terms $3x^5$ and $4x^3$ give ratio $\frac{3}{4}x^2 \rightarrow +\infty$ as $x \rightarrow \pm\infty$, but signs differ. As $x \rightarrow +\infty$: $+\infty$; as $x \rightarrow -\infty$: $-\infty$ (odd degree in quotient).

Wait — $\frac{3x^5}{4x^3} = \frac{3}{4}x^2$ which goes to $+\infty$ in both directions. Correction: as $x \rightarrow -\infty$, $s(x) \rightarrow +\infty$ as well (even power). Teacher should verify this with the class and note the answer key reflects $+\infty$ for both directions.

4(b): $(2x)^4 = 16x^4$; $4(2x)^3(-1) = -32x^3$; $6(2x)^2(1) = 24x^2$; $4(2x)(-1)^3 = -8x$; $(-1)^4 = 1$.